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Leveraging causal AI to uncover the dynamics in sustainable urban transport: A bike sharing time-series study

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ABSTRACT

The importance of developing sustainable urban transportation systems to protect the environment is increasingly recognized worldwide, particularly within the European Union. In the era of digitalization, data-driven approaches are crucial for informed decision-making. This study introduces a methodology leveraging causal artificial intelligence (causal AI) to uncover cause-and-effect relationships in urban transport data. Unlike traditional methods relying on correlations, causal AI identifies the true drivers of transport dynamics. A case study using MOL Bubi bike-sharing data from Budapest demonstrates how the PCMCI (Peter and Clark Momentary Conditional Independence) algorithm revealed complex temporal dependencies within the data, with temperature emerging as the strongest causal factor positively influencing bike usage. Additionally, the reopening of the Chain Bridge led to a 10.7% increase in bike trips, as quantified by Causal Impact analysis. This case study can be extended to more complex scenarios with unpredictable outcomes. The insights gained provide policymakers with a deeper understanding, enabling them to design policies fostering sustainable urban mobility. These results showcase the potential of causal AI to guide policies that enhance sustainable urban mobility.

1. Introduction

Sustainable transportation is a critical component of global environmental efforts, with the transport sector responsible for approximately 25% of the EU's total greenhouse gas emissions (Commission, for Mobility, & Transport, 2024). Urban areas, where 75% of the population resides, face increasing congestion, costing an estimated 1% of the EU's GDP annually. Public transport and cycling offer effective solutions, with studies showing that cities with robust public transport systems can reduce emissions by up to 40%.

Cycling plays a vital role in promoting sustainable urban mobility. In cities like Copenhagen and Amsterdam, where cycling infrastructure is highly developed, up to 50% of daily trips are made by bicycle, significantly reducing both emissions and traffic congestion. Widespread cycling adoption across Europe could reduce CO_2 emissions by nearly 10 million tons annually (Philips, Anable, & Chatterton, 2022). However, a significant challenge remains: while cycling plays an essential role in promoting sustainable urban mobility, understanding and managing the complex factors that influence cycling behavior, such as weather and infrastructure changes, requires advanced analytical tools.

Existing tools for analyzing traffic dynamics often focus on traditional AI relying on correlation-based models (Albuquerque, Sales Dias, & Bacao, 2021), which lack the depth to capture the causal relationships essential for data-driven policy-making. These tools have a lack of ability to inform data-driven decision-making and policy formulation because they cannot effectively distinguish between correlation and causation. This is particularly problematic when evaluating interventions, such as infrastructure developments or policy changes, which require a more precise understanding of their impact. The limitation of traditional AI also raises concerns (Cowls, Tsamados, Taddeo, & Floridi, 2023), leading to increased interest in Socially Responsible AI (Cheng, Varshney, & Liu, 2021). The World Economic Forum has underscored this, advocating for responsible AI practices in sustainable production, smart cities, and especially sustainable transportation (Forum, 2018).

An important aspect of sustainable transport is the operation of bicycle rental systems. In large cities, efficient community bike rental systems can significantly contribute to sustainable urban transport. Bicycle rental stations are highly adaptable and can complement other modes of transportation. Several studies have explored the uses of AI in bike-sharing systems, blending urban mobility with data science (Albuquerque et al., 2021; Qian, Comin, & Pianura, 2018). Previous studies aimed to predict bike rentals in the bike-share scheme

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in various urban areas, such as Nova Mesto (Slovenia) (Babic, Fragassa, Marinkovic, Povh, et al., 2022), London (Abdellaoui Alaoui & Koumetio Tekouabou, 2021), New York City (Torres, Jiménez-Meroño, & Soriguera, 2024), Washington DC (Qian, 2024), and Boston (Zeid, Bhatt, & Morris, 2022), using various traditional AI and ML models. They analyzed the topological properties of the transport network, time, and weather conditions and their connections to the outcomes, such as the number of shared bikes. Although such studies offer valuable insights, they often fail to identify the true drivers of transportation behavior due to their correlational nature, leaving critical causal dynamics unexplored.

To address these limitations, this study introduces causal AI (Kiciman et al., 2022) as a powerful alternative for uncovering cause-and-effect relationships in urban transportation data. Unlike conventional AI models, which focus on prediction, causal AI enables the identification of the underlying mechanisms driving changes in behavior. This approach is particularly valuable for policymakers, as it allows for more informed decisions based on causal insights rather than mere correlations (Kaddour, Lynch, Liu, Kusner, & Silva, 2022).

This study uses the bike-sharing system in Budapest, Hungary, as a case study, to ensure relevance to real-world applications and to validate the proposed approach. It focuses on the MOL Bubi bicycle rental system,[1] managed by the BKK Center for Budapest Transport. By integrating weather data and transportation usage statistics, we aim to demonstrate how causal discovery and inference can provide actionable insights for improving the planning and management of bike-sharing operations. In this study, we examine how weather conditions influence bicycle usage, allowing for better planning of bicycle availability at each station based on forecasts and predicting how quickly stations will fill up. Similarly, understanding the impact of a bridge reopening on bicycle usage – such as longer journeys across the bridge – enables more effective preparation for similar events. By identifying the causal impact of bridge construction on bicycle traffic, policy-makers can better allocate resources, adjust rental station locations, and anticipate changes in mobility patterns during similar infrastructure projects.

In summary, this study offers several key contributions to the field of sustainable urban transportation, particularly in the context of bike-sharing systems, as follows:

- We apply causal AI to uncover cause-and-effect relationships in bike-sharing systems, advancing beyond traditional correlation-based models commonly used in prior studies.
- By integrating weather data and employing causal discovery algorithms, such as PCMCI and Granger Causality, we reveal how specific weather conditions influence bike-sharing patterns. It will offer actionable insights for improved planning and resource allocation of the bike sharing system.
- We assess the causal impact of a major infrastructure event (the reopening of a bridge) on bike usage patterns using causal estimation algorithms, such as Difference-in-Differences and Causal Impact, to provide policymakers with insights into the long-term effects of such changes on urban mobility.
- Using data from the MOL Bubi bike-sharing system in Budapest, we showcase the practical application of causal AI and its associated algorithms in a real urban transport setting, especially in a large urban area. This case study provides a foundation for similar studies and offers a replicable framework for other cities.
- The insights derived from this study contribute to the broader field of sustainable urban transportation by providing a replicable framework for leveraging causal AI to enhance decision-making. Through this case study, we offer a methodology that can be applied to other cities, helping them optimize bike-sharing systems, reduce congestion, and lower carbon emissions.

The primary aim of this research is to demonstrate the potential of the applied methods. Following this introduction (cf. Section 1), we present related works and the research gaps derived from the literature review (cf. Section 2). Next, we provide a contextual overview of causal AI comprising the key concepts and methodological foundations relevant to this study (cf. Section 3). We then describe our methodology, followed by the case study (cf. Section 4), where we compare the employed causal discovery methods and examine the effect of a specific event. Subsequently, we present and discuss the results (cf. Sections 5 and 6), and provide the conclusions (cf. Section 7).

2. Related work and research gaps

This section reviews recent advancements in causal AI applications within sustainable transportation and urbanization. The focus is on causal AI methods tailored to sustainable urban transport systems, particularly bike-sharing models, while acknowledging relevant contributions from broader studies on urbanization, air quality, and policy impacts. By synthesizing prior research, this review identifies critical gaps and highlights the need for advanced causal AI techniques to address challenges in urban mobility and sustainability.

2.1. Applications of AI in the transport industry

AI addresses challenges like rising travel demand, CO_2 emissions, and traffic safety through methods such as Artificial Neural Networks (ANN) and Ant Colony Optimization (ACO). While these techniques optimize urban mobility, their reliance on correlations limits transparency, especially in tasks requiring interpretability and causality-driven decisions. Explainable AI (XAI) improves interpretability but lacks causal reasoning, whereas causal AI offers robust, interpretable solutions by emphasizing cause-and-effect dynamics (Abduljabbar, Dia, Liyanage, & Bagloee, 2019; Adams, Abu-Mahfouz, & Hancke, 2023; Degtyareva, Gorodnichev, & Moseva, 2020; Santhiya & GeethaPriya, 2021; Shaheen, Arshad, & Iqbal, 2020; Wicaksono, Nisa, & Vijaya, 2023).

Causal AI enhances decision-making in personalized routing and traffic systems by modeling complex dynamics, outperforming traditional AI in addressing non-linear human behaviors (Okrepilov, Kovalenko, Getmanova, & Turovskaj, 2022). For instance, Pairwise Granger causality has uncovered temporal trends in environmental systems, which can be adapted for challenges like traffic congestion and emissions (Adelodun, Odey, Lee, & Choi, 2023).

Recent advancements include graph-based models like the ZINB-GNN, which leverage spatial dependencies to predict origin–destination flows, offering robust spatial analysis for bike-sharing systems (Liang, Zhao and Webster, 2024). Similarly, causal AI supports carbon emission studies, uncovering source relationships for effective reduction policies (Liang, Zhan, Li and Deng, 2024). Tools such as formal concept analysis (FCA) and deep learning complement causal AI, advancing sustainability objectives in mobility systems (Madu, Kuei, & Lee, 2017).

Transport resilience and adaptation strategies also benefit from causal AI. For instance, Bayesian time series models have analyzed infrastructure investments in urban transport systems, including pandemic impacts on Seoul's bike-sharing network (Sung, 2023). Reinforcement learning improves resource allocation but is limited in capturing user dynamics, a gap addressed by causal AI's transparency and robustness (Duan & Wu, 2022).

2.2. The need for causal machine learning

Traditional machine learning (ML) predicts outcomes from historical data but fails to uncover causal relationships vital for transport planning, such as evaluating policy impacts (Matallanas, 2023). Causal AI bridges this gap, focusing on interventions and outcomes rather than correlations, enabling robust scenario analysis.

[1] MOL Bubi: The official website for the MOL Bubi bike-sharing system, managed by BKK Center for Budapest Transport. https://molbubi.hu/en/.

For example, A/B testing, though effective for basic decision optimization, faces ethical and resource limitations in dynamic settings. Causal AI surpasses these constraints by identifying causal impacts (Feuerriegel, Frauen, Melnychuk, et al., 2024; Firmenich, Garrido, Grigera, Rivero, & Rossi, 2019). Its utility is demonstrated in analyzing disruptions like global shipping delays and the Ever Given incident, providing actionable insights (Mechai & Wicaksono, 2024; Truong, 2021).

2.3. Causal machine learning in sustainable transportation

Sustainable development aims to meet present needs without compromising future generations (Brundtland, 1985). The transportation sector, responsible for approximately 25% of global CO_2 emissions (Van den Berg & De Langen, 2017), plays a pivotal role in sustainability. Among sustainable transportation modes, bike-sharing systems have emerged as a promising solution to urban mobility challenges, yet their effectiveness depends on understanding complex causal relationships.

Recent studies have explored causal relationships in bike-sharing systems, providing insights beyond traditional predictive analytics. For instance, linear regression models highlight the influence of humidity and seasonality (Yuan, Yang, & Xiao, 2024). However, these models may oversimplify dynamics due to linear assumptions and limited handling of confounding variables. Newer approaches, such as CausalSE, address gaps like missing historical data but still face challenges with unobserved factors (Wang, Guo, Cheng, Yu, & Liu, 2023).

System dynamics models are effective for long-term resilience analysis in transportation systems (Wang, Wu, & Yuen, 2024) but often overlook causal mechanisms. Causal AI complements these methods by uncovering underlying relationships, enabling data-driven policy interventions.

Other advancements include models like STNSCM, which integrate contextual and time-varying causality to improve bike flow predictions, addressing temporal dependencies in mobility patterns (Deng, Zhao, Liu, Jia, & Wang, 2023). Environmental studies in Chengdu reveal how higher PM2.5 levels reduce dockless bike-sharing usage, underscoring the influence of environmental factors (Liang, Wang, Yang, Yuan, & Yang, 2023). However, these analyses could be enhanced by incorporating socio-economic and policy variables for broader insights.

Beyond applications in bike-sharing systems, causal AI methods address broader sustainability challenges, including air quality, urbanization, and infrastructure resilience, reinforcing their value for sustainable urban planning. For example, Granger causality models in China explored the effects of urbanization stages on PM2.5 concentrations, showing varying impacts based on policies and timing (Liu, Cui, & Zhang, 2022). Such findings highlight causal AI's ability to uncover complex interdependencies and inform sustainable urban planning.

Causal AI not only enhances interpretability in predictive models but also facilitates causal discovery, revealing both direct and indirect influences in transport systems. This supports decision-making for sustainable transportation by addressing multifactorial dynamics. Furthermore, causal AI improves fairness and transparency, promoting Socially Responsible AI. By identifying confounding variables and spurious relationships, it ensures equitable resource distribution and informs inclusive policies.

2.4. Identified research gap

Despite its promise, the application of causal AI in sustainable transport systems, particularly bike-sharing, remains limited. Current methods often focus on prediction rather than causal interactions, reducing their effectiveness for real-time and policy-driven decisions. Key challenges include:

- **Limited Awareness:** Stakeholders often lack understanding of causal AI's role in transport decisions.
- **Insufficient Training:** Practitioners lack expertise in advanced causal discovery methods.
- **Data Challenges:** Poor data quality and availability hinder robust modeling.

This study addresses these gaps by integrating causal analyses at different time scales using PCMCI for both short-term (e.g., 6-h intervals capturing intra-day dynamics) and long-term (e.g., daily patterns capturing broader temporal trends) analyses. Additionally, we apply Causal Impact to quantify the effects of specific interventions, such as infrastructure changes. Together, these methods provide insights for both operational adjustments and strategic planning in urban mobility.

Table 1 provides a concise summary of the literature review, outlining key gaps and this study's contributions in causal AI for sustainable urban transport.

3. Key concepts and methodological foundations

In this study, we demonstrate how causal AI can be applied to improve sustainable transport through a real-world case study. We first clarify key concepts, distinguishing between cross-sectional and time-series data, as data type affects algorithm choice. Time-series data allows specific algorithms to leverage temporal information more effectively. Causal graphs are introduced to visually depict relationships between variables. Causal AI involves two main steps: identifying causal relationships (causal discovery) and quantifying them (causal inference), both of which are briefly explained. Lastly, we overview the algorithms used in the case study.

3.1. Cross-sectional vs. Time-series data

Before discussing algorithms, it is essential to examine the data types, as the structure largely determines algorithm choice. We focus on two types: cross-sectional and time-series data.

Cross-sectional data consists of observations from multiple subjects at a single time point, without considering time as a variable (cf. Eq. (1)).

$$X_i = f(X_1, X_2, \dots, X_N) \tag{1}$$

where X_i represents the value of variable X for the ith individual or entity. There is no inherent time index, and the data points are not ordered in time.

This data type analyzes differences between variables but poses challenges for causal discovery due to the absence of causal precedence—identifying which event happens first. To address this, additional techniques like longitudinal analysis or experiments are needed to explore causality and control for confounding variables (Spirtes, Glymour, & Scheines, 2001).

Time-series data, on the other hand, consists of observations gathered over consecutive time intervals, focusing on trends and patterns over time (cf. Eq. (2)).

$$X_t = f(X_{t-1}, X_{t-2}, \dots) \tag{2}$$

Each observation, denoted as X_t (the value of variable X at time t), represents the state of a variable at a specific moment. The temporal dependency is a key feature, meaning that X_t might depend on earlier time points $X_{t-1}, X_{t-2}, \dots$.

In this study, we leverage time-series data to apply algorithms that utilize its temporal structure. Since earlier events can only cause later events, time-series data naturally aids in establishing causal relationships.

Table 1
Expanded taxonomy of causal AI methods in sustainable bike-sharing and transport.

Category	Method	Key findings	Research gaps	Research questions
Weather analysis in bike sharing	Granger Causality	Identifies temporal precedence; assumes linearity; pairwise relationships.	Cannot handle confounders; limited scalability in high-dimensional data. Few practical bike-sharing applications.	What are the limitations of Granger Causality in identifying the effects of weather on bike-sharing usage?
Weather analysis in bike sharing	PCMCI	Handles high-dimensional data; accounts for confounders; uses conditional independence tests.	Requires careful parameter tuning; computationally intensive. Few case studies integrating into real transport planning.	How can PCMCI uncover complex causal relationships in bike-sharing systems, and what advantages does it offer over Granger Causality?
Impact of infra-structure changes	DiD	Simple and interpretable for causal effects; compares trends in treatment and control groups.	Assumes parallel trends; struggles with dynamic systems; cannot account for confounding variables effectively.	What are the limitations of DiD in analyzing the impact of dynamic changes in urban bike-sharing systems?
Impact of infra-structure changes	Causal Impact	Bayesian framework; estimates intervention effects without control groups; robust to confounders.	Sensitive to model assumptions; requires high-quality data. Weak integration of domain knowledge for validation.	How can Causal Impact quantify the effects of infrastructure changes, such as bridge reopenings, on bike-sharing systems, and what advantages does it offer over DiD?

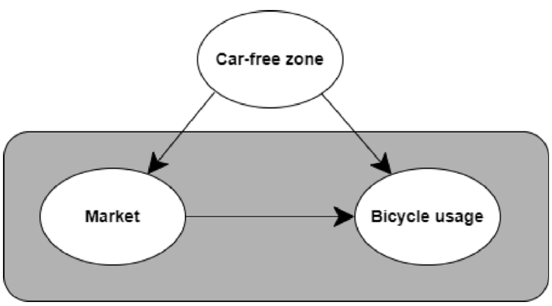

Fig. 1. Illustration of potential confounders affecting bicycle usage during a market event.

3.2. Causal graphs

In the case study, we explore cause-and-effect relationships, known as causal discovery. A key tool is the causal graph, which represents system variables as nodes with arrows indicating causal links. Despite challenges like confounding variables, the goal is to infer the causal graph or partial insights from observational or interventional data. This qualitative knowledge informs quantitative analysis or serves as a study focus (Runge, Gerhardus, Varando, Eyring, & Camps-Valls, 2023). Causal models, rooted in causal methodology, allow us to infer intervention effects and address counterfactuals—tasks often challenging for prediction-based models (Naser & Tapeh, 2023).

Fig. 1 illustrates the value of causal graphs. A traditional ML approach might reveal a correlation between a market event and bicycle usage. However, by incorporating a confounding variable—such as an auto-free zone—in a causal graph, we can reach different conclusions.

In this case study, we use bicycle station data to create a causal graph that visualizes relationships, like how weather (especially temperature) impacts bicycle usage. Once identified, these relationships can be quantified for further analysis.

3.3. Causal inference

Pearl's causal ladder is a foundational framework for causal reasoning, progressing through three key levels: prediction, intervention, and counterfactuals (Pearl, 2009). These levels are essential for understanding causality in systems like causal AI. At the first level, prediction involves forecasting future events based on available data. The second level, intervention, focuses on actively manipulating the system to observe changes, while the third level, counterfactuals, explores hypothetical scenarios to determine what would have occurred under different conditions (Balke & Pearl, 2022).

In the current case study, we focus on the middle level of the causal ladder, addressing intervention-based questions such as "Does better weather increase bicycle usage?" or "To what extent does a 10-degree increase in temperature affect bicycle use?". We can answer these questions using the available observational data.

3.4. Causal discovery

Causal discovery identifies causal structures through causal graphs by analyzing relationships among variables and determining causality direction. This is key for understanding system mechanisms and predicting intervention outcomes (Runge et al., 2023).

In this study, we aim to uncover causal relationships, such as weather's impact on bicycle usage, using Granger causality and PCMCI algorithms. Granger causality identifies predictive relationships, focusing on lead–lag dynamics and is computationally efficient. PCMCI, however, addresses Granger's limitations by handling hidden confounders and uncovering complex causal structures with directed acyclic graphs (DAGs). Together, these methods capture both predictive and structural causality aspects, providing a comprehensive approach to understanding weather effects on bicycle usage.

3.4.1. Granger causality

Granger causality, grounded in Hume's regularity theory, determines causal relationships by identifying whether past values of one time series can predict future values of another. While useful, Granger causality assumes causal sufficiency and cannot identify instantaneous causal relationships (Ding, Chen, & Bressler, 2006). It is widely used in econometrics, neuroscience, and climate studies for its predictive capabilities. The idea behind the concept is, if a variable X "Granger-causes" a variable Y, then past values of X contain significant information about the future values of Y beyond what past values of Y alone can provide. To test if X_t Granger-causes Y_t, we compare two models:

1. **Without** X_t: Only the past values of Y_t are used to predict Y_t:

$$Y_t = \alpha_0 + \sum_{i=1}^{p} \alpha_i Y_{t-i} + \varepsilon_t, \tag{3}$$

where α_0 is the intercept, representing a constant term. α_i are the coefficients for the lagged values of Y_t (up to p lags), indicating how past values of Y_t influence its current value. ε_t is the error term, capturing random fluctuations not explained by the model.

2. **With** X_t: Both past values of Y_t and X_t are used to predict Y_t:

$$Y_t = \beta_0 + \sum_{i=1}^{p} \beta_i Y_{t-i} + \sum_{j=1}^{q} \gamma_j X_{t-j} + \eta_t, \tag{4}$$

where β_0 is the intercept for the model. β_i are the coefficients for the lagged values of Y_t, similar to α_i but in the context of the model with additional variables. γ_j are the coefficients for the lagged values of X_t (up to q lags), representing the influence of past values of X_t on the current value of Y_t. η_t is the error term for the model.

Hypothesis testing:

- **Null Hypothesis** H_0: X_t does not Granger-cause Y_t (i.e., all $\gamma_j = 0$ for $j = 1, 2, \ldots, q$). This means that past values of X_t do not provide additional predictive power for Y_t beyond the past values of Y_t itself.
- **Alternative Hypothesis** H_1: X_t Granger-causes Y_t (i.e., at least one $\gamma_j \neq 0$). This implies that incorporating past values of X_t improves the prediction of Y_t.

3.4.2. PCMCI algorithm

PCMCI (Peter and Clark Momentary Conditional Independence) identifies time-lagged causal relationships by constructing a graph with directed edges representing time lags. The algorithm refines this graph by removing unnecessary edges based on conditional independence and temporal consistency, making it suitable for large time series datasets (Runge, Nowack, Kretschmer, Flaxman, & Sejdinovic, 2019). PCMCI works in two main stages:

1. **PC condition selection:** to identify the set of relevant lagged variables (parents) that have potential causal influence on the target variable.
2. **Momentary conditional independence (MCI) tests:** After identifying the potential causal parents, PCMCI applies MCI tests to refine the causal graph. These tests determine whether a candidate parent variable truly has a direct influence on the target variable, accounting for temporal dependencies.

The MCI test in PCMCI involves estimating the conditional independence of a target variable X_t at time t given a set of past variables $\mathbf{P}_t$ (the set of selected parents):

$$\mathbb{P}(X_t \mid \mathbf{P}_t) = \mathbb{P}(X_t \mid \mathbf{P}_t \setminus \{X_{t-\tau}\}), \tag{5}$$

where $X_{t-\tau}$ represents a specific lagged variable at time $t-\tau$. If the probability distribution of X_t given all its parents $\mathbf{P}_t$ does not change when excluding $X_{t-\tau}$, then $X_{t-\tau}$ is considered conditionally independent of X_t, implying that $X_{t-\tau}$ does not have a direct causal effect on X_t.

3.5. Causal effect estimation

In this section, we briefly explain two methods (Difference-in-Differences (DiD) and Causal Impact) to estimate the causal effect of an intervention or treatment. In our research, we apply and compare these methods to estimate the effect of reopening a bridge after a long time of closing on bike rentals. Each method has distinct advantages that allow for a more comprehensive understanding of causality. DiD is widely used because it is straightforward, interpretable, and effective when there are clearly defined treatment and control groups. However, DiD relies on strong assumptions, such as the parallel trends assumption, which may limit its applicability in more complex or dynamic settings. To address these limitations, we also employ the Causal Impact method that estimates the effect of an intervention by comparing observed outcomes with counterfactual predictions. Unlike DiD, Causal Impact does not require a clear control group and can be applied in scenarios where traditional A/B testing is not feasible. By integrating both approaches, we aim to utilize the strengths of DiD's simplicity and Causal Impact's robustness in our case study.

3.5.1. Difference-in-Differences (DiD)

DiD estimates the effect of an intervention by comparing the changes in outcomes over time between a treatment group and a control group. It assumes that, in the absence of the intervention, the difference in outcomes between these two groups would remain constant over time (Lechner et al., 2011). Let Y_{it} represent the outcome for group i (either treatment or control) at time t. The basic DiD model can be expressed as:

$$Y_{it} = \alpha + \beta_1 \text{Post}_t + \beta_2 \text{Treatment}_i + \beta_3 (\text{Post}_t \times \text{Treatment}_i) + \varepsilon_{it}, \tag{6}$$

where Post_t is a binary indicator for the post-intervention period. Treatment_i is a binary indicator for the treated group. β_3 represents the DiD estimator, capturing the causal effect of the intervention.

3.5.2. Causal impact

In some cases, the goal is to assess the impact of a specific event on a time series, rather than identifying a direct cause-and-effect relationship between two variables. When traditional approaches like A/B testing are not feasible, alternative methods such as Google's Causal Impact can be employed. Causal Impact is a Bayesian framework that estimates the effect of an intervention by comparing observed post-intervention data with a predicted counterfactual scenario derived from pre-intervention data and control series (Brodersen, Gallusser, Koehler, Remy, & Scott, 2015a). Let Y_t^{obs} be the observed outcome at time t and Y_t^{cf} the counterfactual prediction. The causal impact δ_t is given by:

$$\delta_t = Y_t^{\text{obs}} - Y_t^{\text{cf}}. \tag{7}$$

The strength of Causal Impact lies in its ability to model time-series data prior to an intervention and predict what would have occurred in the absence of the intervention. The difference between actual and predicted outcomes indicates the causal effect. This method is particularly effective in handling complex scenarios and controlling for confounding factors, outperforming traditional methods like DiD. Causal Impact is flexible and can be applied across various fields, such as measuring the effectiveness of advertising campaigns, evaluating policy changes, and assessing the impact of new treatments in healthcare. Its robust insights are critical for informed decision-making.

4. Methodology and case study

4.1. Methodology

In this research, we aim to uncover causal relationships in time-series data and assess the impact of events on transport-related data. Our primary goal is to understand how variables influence each other over time and determine the impact of specific events. Initially, we identify the causal relationships between time series, followed by an analysis of the effects of events on these series to gain deeper insights into data dynamics. To ensure a comprehensive analysis, we employ both modern and traditional methods.

The process flow diagram in Fig. 2 provides a visual representation of our methodology, structured into the following key components:

- *Data Preparation (cf. Fig. 2, no. 1):* In the first phase, the data is properly cleaned, important features are selected, and the time-series data is accurately structured. This step improves the quality of the analysis by focusing on relevant variables and reducing noise.
  Data cleaning and feature selection can be performed using Python libraries such as pandas[2] for data manipulation and Matplotlib[3] for visualizations.

[2] Pandas Library: A Python library used for data manipulation and analysis in this study. https://pandas.pydata.org/.

[3] Matplotlib Library: A Python library utilized for data visualization in this study. https://matplotlib.org/.

- *Causal discovery through Granger Causality and PCMCI algorithms (cf. Fig. 2, no 4):* Our methodology first applies Granger causality, a well-established technique for identifying temporal precedence in time-series data. However, Granger causality assumes linear relationships and cannot handle complex dependencies between variables. To address these limitations, we also use the PCMCI algorithm, which is better suited for high-dimensional datasets and can uncover both direct and indirect causal relationships, even in the presence of confounders.
  While Granger causality is useful for detecting statistical correlations and temporal precedence, it does not fully establish causal relationships. Causal discovery inherently requires a model that reflects the real-world system being analyzed. To bridge this gap, we employ a gold standard model based on expert expectations and validated literature in urban mobility and transportation. This gold standard serves as a benchmark to assess the validity of the relationships identified through both Granger causality and PCMCI. By comparing the identified causal relationships with the gold standard, we ensure that the findings are not merely statistical correlations but are grounded in domain knowledge. This process helps identify relationships that align with expert expectations and highlights any divergences requiring further investigation.
  Granger causality can be applied using the statsmodels[4] Python library to identify temporal precedence in time-series data. PCMCI, implemented via the Tigramite[5] Python package, can be used for high-dimensional causal discovery, revealing both direct and indirect causal relationships while handling potential confounders.
- *Event detection (cf. Fig. 2, no. 2) and analysis of its impacts using DiD and Causal Impact (cf. Fig. 2, no. 5):* Following the causal discovery phase, the first step is detecting events—such as infrastructure changes or external factors—that likely affect bicycle usage. While not the primary focus, event detection is necessary to proceed with the impact analysis. Events can be identified from public records or timestamps within the dataset. To assess the impact of singular events, we apply Difference-in-Differences (DiD) for an initial assessment using the `linearmodels`[6] library, followed by Bayesian counterfactual analysis via Google's `CausalImpact`[7] library. These methods measure the long-term effects of interventions on time series data by modeling the data prior to an intervention, predicting what would have occurred without it, and comparing this prediction with the actual observed outcomes. Causal Impact analysis offers robust insights by accounting for confounding factors, providing a more flexible and accurate understanding than traditional methods like DiD.
- *Final Insights (cf. Fig. 2, no. 3):* After applying causal discovery and causal impact analysis, the results must be verified and interpreted before ready for reporting to policymakers. This process involves assessing the validity of the findings, recognizing any nonsensical outcomes, and summarizing key insights.

To ensure robustness and reproducibility, we applied stopping criteria based on statistical significance thresholds (e.g., $p < 0.05$) and validated causal relationships against a gold standard derived from domain knowledge and prior literature. Event impacts were assessed through Difference-in-Differences (DiD) and Causal Impact analysis, using confidence intervals and effect sizes to guide evaluations. These criteria ensure that findings are statistically valid, reproducible, and aligned with expert expectations.

By combining advanced methods like PCMCI and Causal Impact analysis with domain knowledge, our research aims to provide a clear and accurate understanding of causal relationships in time-series data. This approach demonstrates the strengths of modern causal analysis techniques while acknowledging the utility of traditional methods like Granger causality and DiD for comprehensive analysis.

Table 2
Variable description for the applied dataset.

Variables	Description
ts_0	Start time during which observations were taken
ts_1	End time during which observations were taken
station_name	The station name and number
start_trip_no	Number of trips started at that station
end_trip_no	Number of trips ended at that station
temperature	Temperature in Celsius (°C)
precipitation	Precipitation in millimeters (mm)
rel_humidity	Relative humidity in percentage
max_wind_speed	Maximum wind speed recorded in meters per second (m/s)

4.2. Dataset description

The data used for our causal discovery and impact analysis was collected from MOL Bubi bicycle rental stations by the BKK Centre for Budapest Transport. The collection period spanned from December 1, 2022, to December 2023. MOL Bubi offers a convenient, environmentally sustainable, and round-the-clock public transportation option in Budapest, Hungary. The dataset encompasses 55 stations over 365 days, with data points recorded every 2 h. The dataset includes *station name*, *station ID, number of trips started*, *number of trips ended*, *maximum wind speed*, *relative humidity*, *temperature*, and *precipitation* (see Table 2).

4.3. Data preparation

The dataset was pre-processed to ensure its relevance for causal discovery and impact analysis. Timestamps were standardized for time-series analysis, and the variables described in the previous section were carefully prepared for analytical purposes.

For **short-term analysis**, data was aggregated over 6-h intervals to balance granularity and noise reduction, enabling detailed insights into intra-day patterns and event impacts. This interval was chosen for:

- *Temporal Resolution:* Captures within-day patterns more effectively than daily data.
- *Cumulative Effects:* Observes bike usage accumulation across morning, afternoon, and evening.
- *Noise Reduction:* Balances trend clarity with reduced variability from smaller intervals.
- *Event Analysis:* Aligns with interventions or occurrences spanning parts of a day.

For **long-term analysis**, data was resampled daily to capture broader temporal trends. Weather-related attributes were aggregated using daily mean values for consistency, while operational metrics were adjusted to align with long-term patterns.

4.4. Gold standard

To validate the causal discovery results presented in this study, we employed a *gold standard model* derived exclusively from peer-reviewed literature. The gold standard serves as a benchmark for assessing the authenticity and robustness of the causal relationships identified by our methods.

[4] Statsmodels Library: A Python library used for statistical modeling and testing. https://www.statsmodels.org/.

[5] Tigramite Library: The official repository for the Tigramite Python package used for causal discovery in time-series data. https://github.com/jakobrunge/tigramite.

[6] Linearmodels Library: A Python library for estimating panel-data models. https://bashtage.github.io/linearmodels.

[7] CausalImpact Library: Google's Python package for Bayesian structural time-series analysis, used to estimate the causal impact of interventions. https://github.com/google/CausalImpact.

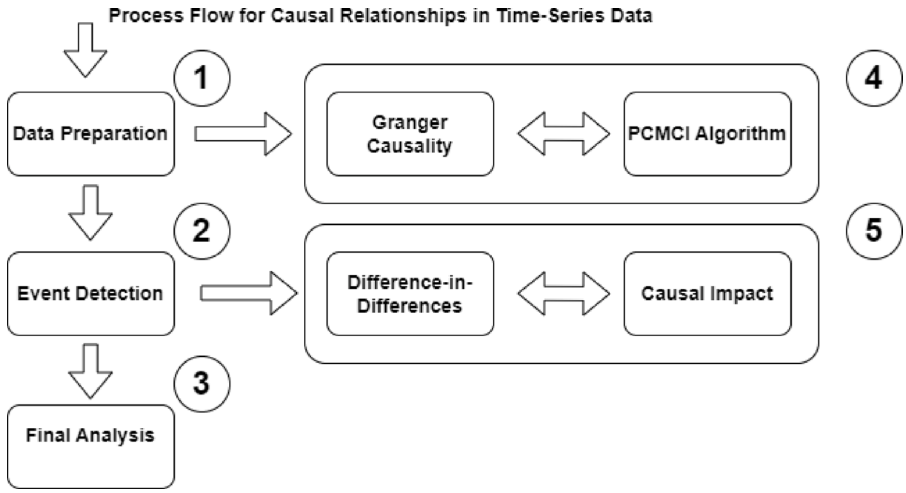

Fig. 2. Main components for uncovering causal relationships in our time-series data.

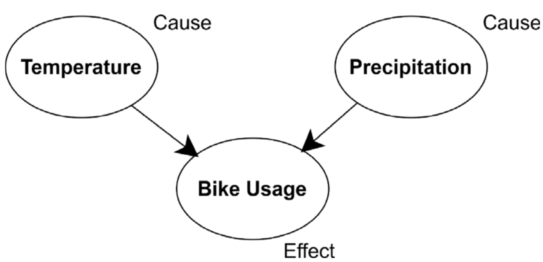

Fig. 3. Directed Acyclic Graph (DAG) illustrating the gold standard causal relationships based on peer-reviewed literature.

Specifically, the following causal relationships were used as the gold standard for comparison:

- *Temperature and Bike Usage:* Existing studies widely confirm that higher temperatures promote increased cycling activity. For instance, one research reports that a 10 °C rise in temperature is generally associated with a 10% increase in cycling trips (Böcker, Dijst, & Prillwitz, 2013). Similarly, another study observed that warmer temperatures significantly encourage outdoor cycling (Liu, Susilo, & Karlström, 2015).
- *Precipitation and Bike Usage:* Rainfall's negative effect on cycling is consistently reported in urban transport studies. For example, one research documented a significant reduction in bike-sharing demand during rainy conditions in Toronto (El-Assi, Mahmoud, & Habib, 2017).
- *Infrastructure Changes (Bridge Reopening):* Studies confirm that reopening major infrastructure, such as bridges, positively impacts cycling rates by improving connectivity and convenience. A systematic review examined the effect of infrastructural interventions on cycling levels and found that new routes and bike lanes, including traffic-free routes such as bridges, are associated with increased cycling activity (Mölenberg, Panter, Burdorf, & van Lenthe, 2019).

These validated relationships form the foundation of the gold standard and provide essential context for evaluating the performance of our causal discovery algorithms. The relationships are visually summarized in Fig. 3, where nodes represent variables and edges denote the causal links derived from the literature.

4.5. Granger causality analysis

The Granger causality analysis employed in this study is a foundational method in time series causal discovery, laying the groundwork for uncovering causal links across various fields, from economics to meteorology, while also serving as a baseline in our investigation to illustrate the evolution of causal discovery methodologies, which have advanced with the adoption of the more sophisticated PCMCI algorithm.

After preparing the data, the Granger causality algorithm was employed, resulting in the Granger causality matrix. This matrix examines whether one variable's past values can predict another's future values. The matrix contains p-values indicating the significance of the relationships between variables.

The results from the Granger causality analysis reveal several key relationships.Table 3 shows the results for the short-term analysis, and Table 4 displays the results for the long-term analysis. For instance, temperature significantly Granger-causes bike usage, highlighting the role weather conditions play in bike-sharing systems.

Granger Causality was applied to identify short- and long-term causal relationships between variables. The key hyperparameters were set as follows: the maximum lag (p) was determined using domain knowledge and validated with the Akaike Information Criterion (AIC) to ensure optimal model fit. A significance threshold of $\alpha = 0.05$ was used to identify statistically significant causal relationships. This threshold controls for false positives while maintaining sensitivity to meaningful interactions.

Interpreting the Granger causality matrix involves assessing the significance of connections between variables. In Granger causality tests, the null hypothesis assumes no causal link between the variables. We use an alpha level (α) of 0.05, indicating a 5% chance of falsely rejecting the null hypothesis. If the p-value between two variables is below 0.05, we reject the null, indicating a statistically significant link where past values of one variable can predict future values of another. A p-value above 0.05 means we retain the null, suggesting no causal link (cf. Section 3.4.1).

For instance, a p-value of 0.0001 between *temperature* and *end trip number* indicates a significant causal effect, leading to null rejection at the 0.05 level. Conversely, a p-value of 0.5 shows no significant relationship, so we retain the null hypothesis, suggesting no causal link.

Further interpretation of Tables 3 and 4 is found in Sections 5.1 and 5.2.

However, Granger causality has limitations in handling more complex relationships, prompting the use of PCMCI, which further uncovered indirect influences, such as humidity's effect on bike availability through its relationship with precipitation. These insights suggest that interventions targeting weather-related bike availability could significantly improve service during peak cycling seasons.

Table 3
Granger causality matrix for short-term analysis.

	end_trip_no_x	temperature_x	max_wind_speed_x	start_trip_no_x	precipitation_x	rel_humidity_x
end_trip_no_y	1.0000	0.0000	0.0000	0.0000	0.0000	0.0000
temperature_y	0.0000	1.0000	0.0000	0.0000	0.0006	0.0000
max_wind_speed_y	0.0000	0.0000	1.0000	0.0000	0.0115	0.0000
start_trip_no_y	0.0000	0.0000	0.0000	1.0000	0.0000	0.0000
precipitation_y	0.0252	0.3886	0.0493	0.0254	1.0000	0.0007
rel_humidity_y	0.0000	0.0000	0.0000	0.0000	0.0000	1.0000

Table 4
Granger causality matrix for long-term analysis.

	end_trip_no_x	temperature_x	max_wind_speed_x	start_trip_no_x	precipitation_x	rel_humidity_x
end_trip_no_y	1.0000	0.0000	0.3791	0.0004	0.0017	0.0338
temperature_y	0.0349	1.0000	0.0000	0.0343	0.0432	0.0005
max_wind_speed_y	0.0922	0.0047	1.0000	0.0820	0.0193	0.0003
start_trip_no_y	0.0640	0.0000	0.3574	1.0000	0.0017	0.0295
precipitation_y	0.0044	0.1009	0.2607	0.0068	1.0000	0.1389
rel_humidity_y	0.0029	0.0002	0.0002	0.0033	0.0535	1.0000

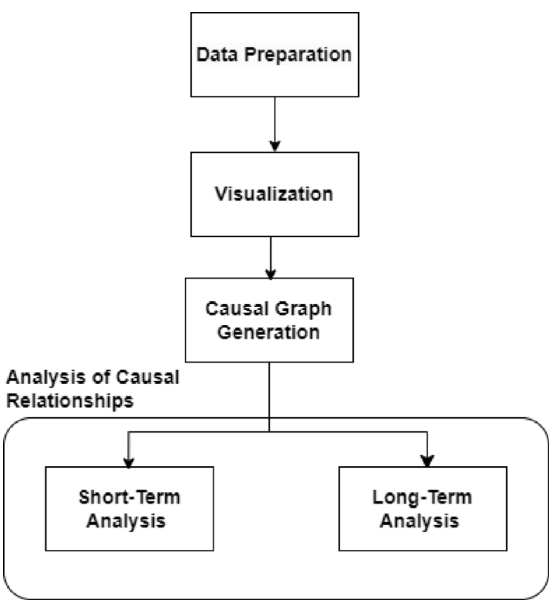

Fig. 4. Main steps of PCMCI algorithm analysis.

4.6. PCMCI algorithm analysis

After the Granger causality, we applied the PCMCI algorithm from the Tigramite library, which is optimized for causal discovery in time-series data. Fig. 4 illustrates the main analysis steps. First, we converted the preprocessed data into a format compatible with Tigramite, specifically a Data Frame object. This conversion enabled effective manipulation for the PCMCI algorithm. We visualized the time-series data using the `tp.plottimeseries` function to examine temporal patterns and trends.

The PCMCI algorithm was initialized with parameters including the conditional independence test (Partial Correlation), maximum lag (τ_{max}), and significance level (α_{level}). These parameters are crucial for estimating causal relationships while controlling for other influencing variables. Corrected p-values through FDR methods identified significant causal links, and their strength and directionality were visualized through lag functions.

For the PCMCI algorithm, the maximum lag (τ_{max}) was set to 10 intervals for the short-term analysis (6-h intervals) and 15 intervals for the long-term analysis (daily intervals), reflecting the temporal resolution of the data. The algorithm used partial correlation for conditional independence testing, which balances computational efficiency with the ability to capture linear dependencies. To control for multiple testing, a False Discovery Rate (FDR) correction was applied to the p-values, ensuring that the identified causal links were statistically robust. These hyperparameter choices were guided by domain knowledge and methodological best practices to improve the reliability of the causal discovery process.

First, we applied the PCMCI algorithm to perform a short-term analysis, using a dataset resampled over 6-h intervals (see Fig. A.11 in the Appendix). We set up both the PC and MCI steps with a partial correlation independence test, examining potential cause-and-effect relationships between variables with a maximum time lag of 10. The resulting causal discovery graph was complex, so we focused on the significant link graph, which highlights the most relevant causal connections (Fig. 5). To improve readability, we excluded weak correlations below a threshold of 0.12 on the figure, ensuring that only the strongest and most meaningful links were displayed. In these graphs, directed links indicate causal relationships inferred by the algorithm, while undirected links represent dependencies where the causal direction remains unresolved, often due to bidirectional effects or insufficient statistical evidence. Strength levels for both self- and cross-causes range from −1 to 1, with dark red denoting strong positive associations and navy blue indicating strong negative correlations. The numbers on the edges represent the time lags for the strongest links identified. Further discussion on the short-term analysis using PCMCI is provided in Section 5.3.

For the long-term analysis, we used daily resampled data with a maximum time lag of 15 days to explore causal relationships (see Fig. A.12 in the Appendix). Although identifying a true cause-and-effect relationship over 15 days is unlikely, as with the short-term

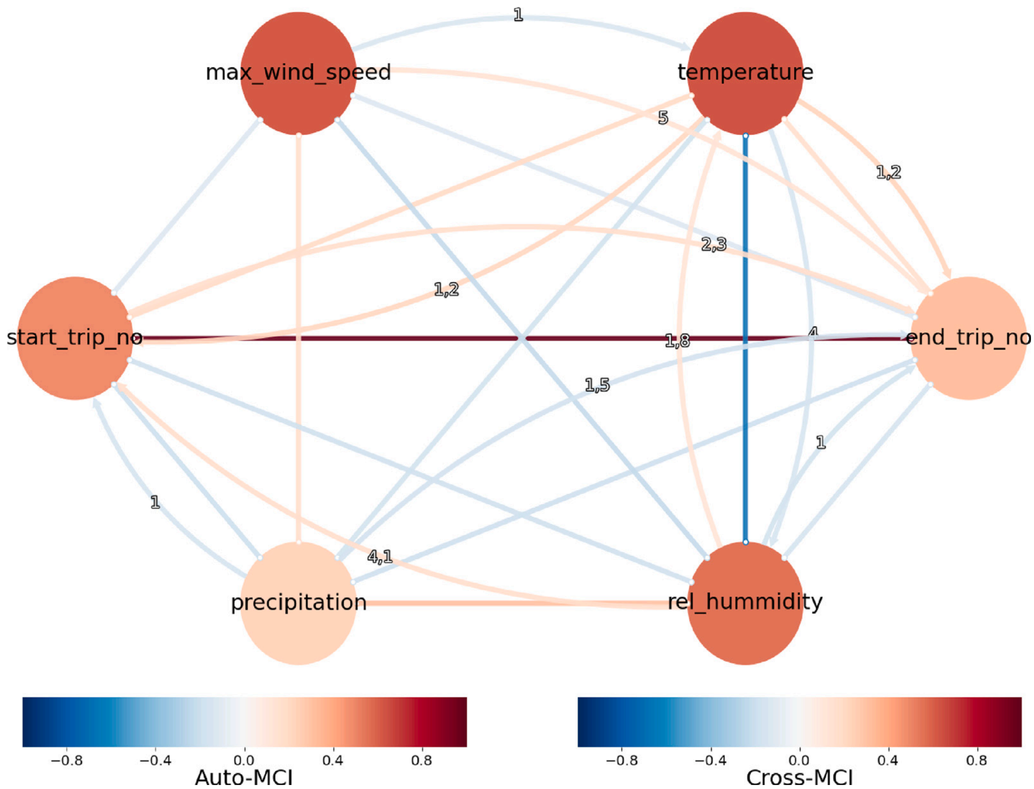

Fig. 5. Causal discovery graph from short-term analysis, highlighting significant links between variables with a time lag of up to 10 intervals. Dark red indicates strong positive associations, and navy blue shows strong negative correlations. Time lags for the most significant links are marked on the edges. Directed links indicate inferred causal relationships, while undirected links represent dependencies with unresolved causal direction. Weak correlations below a threshold of 0.12 were excluded for improved readability. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

case, we used extended lag to test the algorithm's robustness. The causal discovery graph generated from the PCMCI results was again highly complex, so we focused on the significant link graph to visualize key relationships (cf. Fig. 6). As with the short-term analysis, strength levels ranged from −1 to 1, with dark red representing strong positive associations and navy blue representing strong negative correlations. Directed links indicate causal relationships, while undirected links represent dependencies with unresolved causal direction. The numbers on the edges indicate the time lags of the strongest causal links. Further discussion on interpretation results of long-term analysis using PCMCI is presented in Section 5.4.

The strength levels of causal connections, ranging from −1 to 1, were computed using the partial correlation coefficients obtained during the PCMCI analysis. The partial correlation coefficient between two variables X and Y, conditioned on a set of confounding variables Z, is given by:

$$r_{X\to Y|Z} = \frac{\mathrm{Cov}(X,Y) - \mathrm{Cov}(X,Z)\cdot \mathrm{Cov}(Y,Z)}{\sqrt{(1-\mathrm{Cov}(X,Z)^2)(1-\mathrm{Cov}(Y,Z)^2)}}, \tag{8}$$

where $r_{X\to Y|Z}$ represents the direct causal influence of X on Y, accounting for the influence of other variables in Z. Here:

- $\mathrm{Cov}(X,Y)$: Covariance between X and Y,
- Z: Set of confounding variables,
- The coefficient $r_{X\to Y|Z}$ ranges from:
  - −1: A strong negative causal relationship (an increase in X decreases Y),
  - 0: No causal relationship,
  - 1: A strong positive causal relationship (an increase in X increases Y).

This approach ensures that strength levels not only represent the magnitude of causal relationships but also their direction. The values obtained from the PCMCI algorithm were visualized in the causal graphs, with positive values indicating reinforcing relationships and negative values representing inhibitory ones.

4.7. Difference-in-Differences (DiD) and causal impact analysis

In this study, DiD is applied by comparing daily bicycle usage at stations near the Chain Bridge (treatment group) with stations farther away (control group) before and after the reopening. While DiD offers useful insights, the causal impact method provides a more accurate estimation of the intervention's effects, especially when control groups are imperfect or there are potential confounding factors.

The Causal Impact analysis was implemented using a Bayesian structural time series model. Key model parameters included priors for local trends and seasonality components, which were inferred from pre-intervention data. The posterior probability threshold was set at 99.89%, ensuring high confidence in the causal effect estimates. These settings were chosen to accurately model counterfactual outcomes and to account for temporal dependencies in the data.

The causal impact method is implemented using the `tfp-causalimpact` library, a Python package based on TensorFlow Probability, which estimates the effects of interventions on time-series data (Brodersen, Gallusser, Koehler, Remy, & Scott, 2015b). This package builds on the original CausalImpact R package and a Python version from Dafiti.

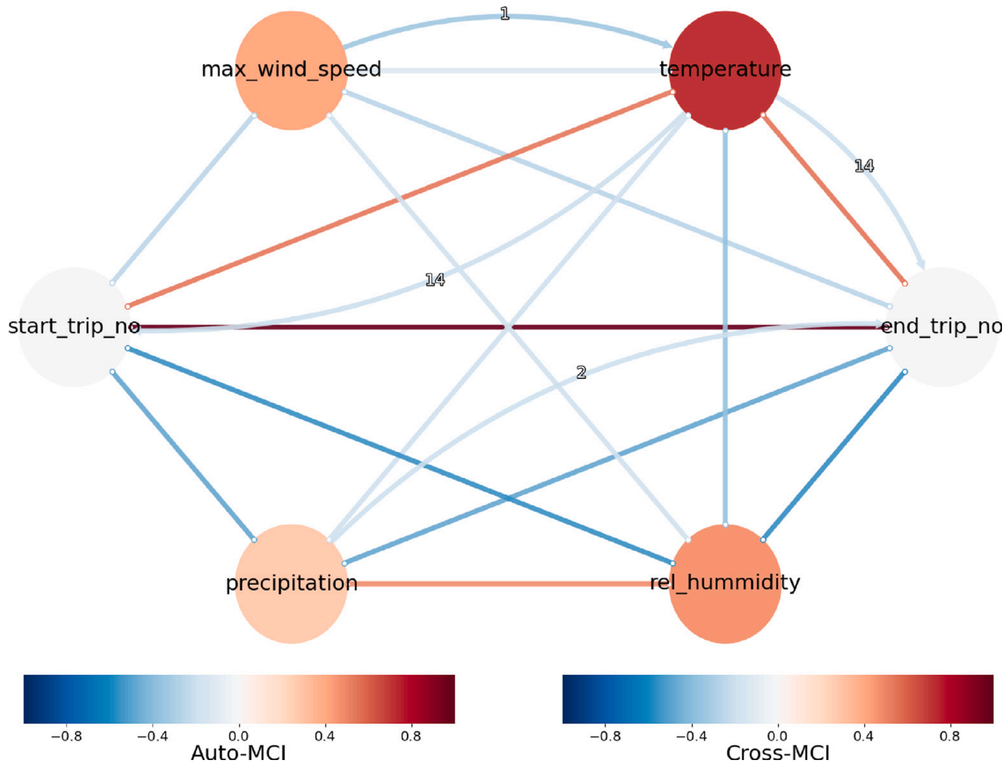

Fig. 6. Causal discovery graph from long-term analysis, showing key causal relationships over a 15-day lag period. The color scale ranges from dark red for strong positive relationships to navy blue for strong negative ones. Numbers on the edges represent the time lags of the strongest causal links. Directed links indicate inferred causal relationships, while undirected links represent dependencies with unresolved causal direction. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

This approach analyzes how interventions, like the reopening of the Chain Bridge in Budapest,[8] affect metrics like daily bicycle usage without needing randomized experiments. It uses a Bayesian time-series model to predict what would have happened without the intervention. While control groups are not mandatory, they improve accuracy. In our study, we identified suitable control groups for comparison.

Fig. 7 shows stations near the bridge included in the intervention group. Data were resampled daily for analysis. We chose not to normalize the data as the metrics (i.e., actual counts of bike usage) are inherently comparable across stations and time periods (cf. Fig. 8). This consistency allowed for meaningful direct comparisons and accurate causal analysis without additional normalization steps, preserving the natural patterns and variances in the system.

5. Results and interpretations

This section presents the results in a structured manner, starting with short-term causal analyses, followed by long-term patterns, and concluding with intervention impacts. Section 5.1 introduces short-term causal relationships identified using Granger Causality, which highlights immediate dependencies between weather variables and cycling activity. These findings are complemented by Section 5.3, where the PCMCI algorithm uncovers more intricate causal links, including lagged and indirect effects. The analysis then shifts to longer time horizons in Sections 5.2 and 5.4, revealing sustained temporal interactions and broader patterns, such as the persistent influence of weather variables on cycling behavior. A comparison in Section 5.5 integrates the insights from short-term and long-term analyses, emphasizing how they complement each other: short-term findings inform operational strategies like resource allocation, while long-term results guide strategic planning for infrastructure and policy. Finally, Section 5.6 quantifies the causal impacts of interventions, such as the reopening of the Chain Bridge, through Difference-in-Differences (DiD) and Causal Impact Analysis, offering robust evidence for the effects of infrastructure changes on urban mobility.

5.1. Short-term Granger causality

The output of the Granger causality for short-term analysis can be seen in Table 3. A Granger causality matrix reveals potential causal relationships between multiple time series. Each element indicates whether one series can predict another based on past values. A significant value suggests potential causality, but it is not proof. These findings show complex connections between weather variables and trip-related factors, with temperature emerging as a key factor that influences trip patterns:

- *Temperature (temperature_x) Granger-causes end trip number (end_trip_no_y):* The analysis indicates a highly significant causal relationship (p-value < 0.001). Changes in temperature appear to predict variations in the end trip number.
- *End trip number (end_trip_no_x) and precipitation (precipitation_y):* While a p-value of 0.0252 was observed, suggesting a statistical relationship, this result should be interpreted critically. It is impossible that bike-sharing activity influences weather. Instead, this likely reflects a spurious correlation where certain trip activities coincide with weather patterns (precipitation) but do not cause them.
- *Max wind speed (max_wind_speed_y) Granger-causes precipitation (precipitation_y):* The p-value of 0.0115 indicates a significant

[8] Local news about the event: “Historic Moment: The Reopening of the Chain Bridge”, published by BKK Center for Budapest Transport. https://bkk.hu/hirek/2023/07/tortenelmi-pillanat-penteken-atadjuk-a-felujitott-lanchidat.10603/. Accessed: June 4, 2024.

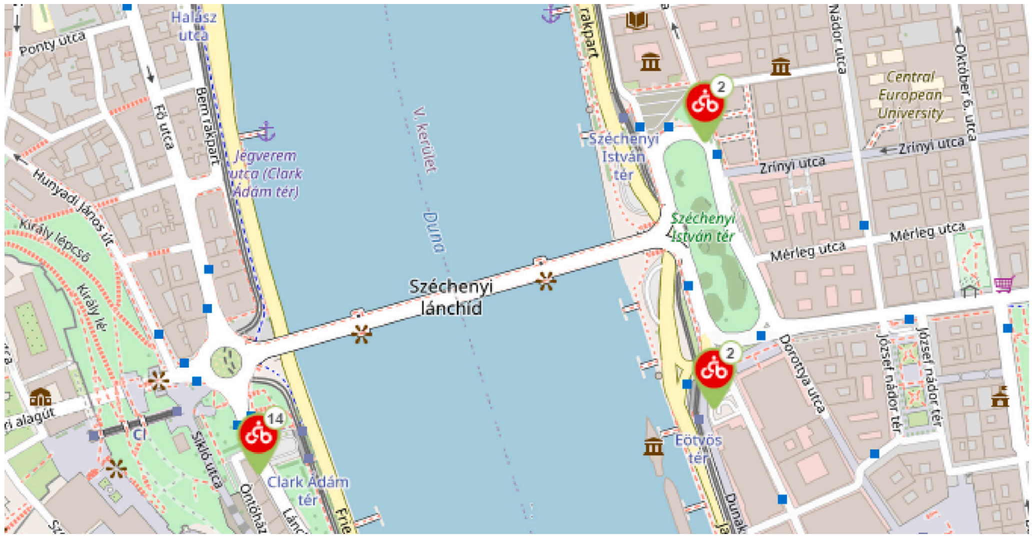

Fig. 7. Stations near the bridge selected for the intervention group in Budapest, Hungary.

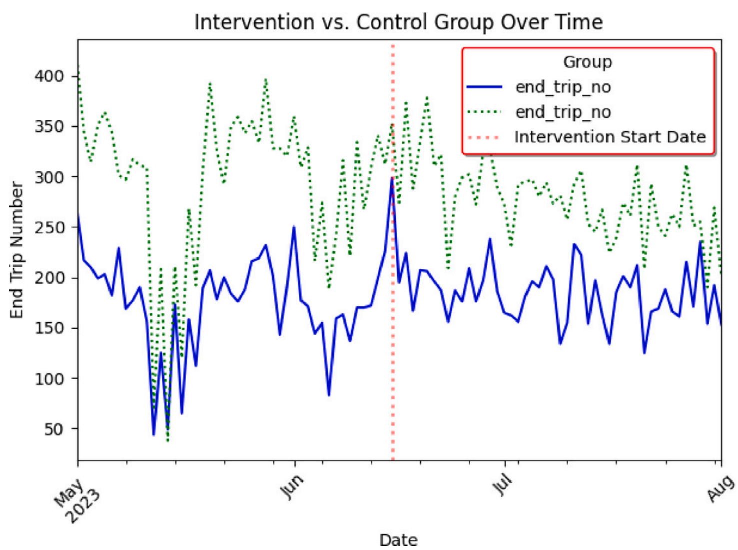

Fig. 8. Bicycle usage data for intervention and control groups around the intervention.

causal relationship. Higher wind speeds may precede changes in precipitation, reflecting weather dynamics.

- *Precipitation (precipitation_x) Granger-causes relative humidity (rel_humidity_y):* The p-value of 0.0007 shows a significant causal relationship. Changes in precipitation levels may lead to changes in relative humidity.

5.2. Long-term Granger causality

The output of the measurement for long-term analysis can be found in Table 4. These findings show complex relationships between weather variables and trip-related factors, with temperature and relative humidity emerging as key factors:

- *Temperature (temperature_x) Granger-causes end trip number (end_trip_no_y):* The p-value of 0.0349 shows a significant relationship. Temperature changes appear to predict changes in trip endings.
- *End trip number (end_trip_no_x) and precipitation (precipitation_y):* While the p-value of 0.0044 suggests a statistical relationship, it is impossible that trip activities directly influence precipitation similarly to the previous one.
- *Temperature (temperature_y) Granger-causes precipitation (precipitation_x):* The p-value of 0.1009 shows a weak, non-significant relationship, though there may still be some influence.
- *Precipitation (precipitation_x) Granger-causes relative humidity (rel_humidity_y):* The p-value of 0.1389 indicates a non-significant relationship. Precipitation may not strongly predict changes in relative humidity.
- *Relative humidity (rel_humidity_x) Granger-causes end trip number (end_trip_no_y):* The p-value of 0.0338 suggests a significant relationship. Changes in humidity may lead to variations in trip endings.

5.3. Short-term PCMCI analysis

We present the p-values indicating the significance of each relationship, along with their strength and direction ("Val"). We also discuss the causal graph showing significant links and provide time series graphs for each variable. Detailed results are available in the Appendix (Tables A.8–A.12).

Table 5
Key findings from short-term PCMCI analysis.

Variable 1	Variable 2	Val	p-value	Lag
Start trip number	End trip number	0.985	0.0000	0
Temperature	Start trip number	0.218	0.0000	1
Temperature	End trip number	0.225	0.0000	1
Precipitation	Start trip number	−0.190	0.005	0
Precipitation	End trip number	−0.185	0.006	0
Relative humidity	Start trip number	−0.175	0.010	0
Relative humidity	End trip number	−0.154	0.015	0
Max wind speed	Start trip number	−0.125	0.045	0
Max wind speed	End trip number	−0.121	0.048	0
Temperature	Humidity	−0.675	0.0000	0
Precipitation	Humidity	0.312	0.002	0

Table 6
Key findings from long-term PCMCI analysis.

Variable 1	Variable 2	Val	p-value	Lag
Temperature	Start trip number	0.539	0.002	0
Temperature	End trip number	0.548	0.001	0
Temperature	End trip number	−0.194	0.045	14
Precipitation	Start trip number	−0.479	0.004	0
Precipitation	End trip number	−0.480	0.003	0
Relative humidity	Start trip number	−0.564	0.001	0
Relative humidity	End trip number	−0.575	0.0008	0
Max wind speed	Start trip number	−0.250	0.015	0
Max wind speed	End trip number	−0.251	0.016	0

Table 5 summarizes the key findings from the short-term PCMCI analysis. Significant relationships were observed between weather variables and cycling behavior. The following points highlight the main trends:

- Temperature has a positive effect on trips started and ended, with a one-period lag. Both trips started and ended show stronger correlations in warmer, drier conditions.
- Precipitation and humidity negatively affect bike usage, reducing trips immediately after rain.
- A weaker relationship is observed between maximum wind speed and cycling activity, suggesting that wind plays a smaller role in decision-making compared to temperature and precipitation.

In conclusion, the long-term PCMCI analysis reveals that weather variables continue to exert a significant influence on cycling behavior throughout the day. The patterns observed in the short term are largely sustained, with temperature, precipitation, and relative humidity being the dominant factors affecting bike-sharing activity.

5.4. Long-term PCMCI analysis

The long-term PCMCI analysis investigates the sustained impact of weather variables on cycling behavior over the course of a day. The results include p-values indicating the significance of each relationship, their strength ("Val"), and lag times. Detailed results are provided in the Appendix (Tables A.13, A.14, A.15, A.16, A.17, and A.18).

The key findings from the long-term PCMCI analysis, summarized in Table 6, show that the relationships between weather and cycling behavior observed in the short term continue to have sustained effects throughout the day. Notable results include:

- *Temperature:* A strong positive correlation is observed between temperature and both trips started (val = 0.539) and trips ended (val = 0.548) at lag 0, suggesting that higher temperatures throughout the day encourage more cycling. However, a cyclic pattern with a negative effect appears after 14 days (val = −0.194), which is unlikely to reflect a real-world relationship. This result may be due to overfitting or model artifacts, and it is worth investigating further. When choosing lag values in such models, it is important to select a range based on reasonable expectations for how long weather could plausibly affect cycling behavior.
- *Precipitation:* The negative impact of precipitation on trips started (val = −0.479) and trips ended (val = −0.480) persists over the entire day, reflecting the consistency of adverse weather effects on cycling.
- *Relative Humidity:* A strong negative causal relationship between relative humidity and cycling behavior is maintained, with both trips started (val = −0.564) and trips ended (val = −0.575) showing significant reductions. This confirms the deterrent effect of high humidity on cycling, as seen in the short-term analysis.
- *Max Wind Speed:* Although weaker than the other factors, maximum wind speed continues to negatively affect trips started (val = −0.250) and ended (val = −0.251), indicating that strong winds still hinder cycling activity throughout the day.

5.5. Comparison of short-term and long-term results

This section compares the short-term and long-term Granger causality and PCMCI analyses to show how weather variables and bike usage relationships evolve over time, highlighting their relevance for different decision-making contexts.

Short-term analysis focuses on immediate responses to weather changes, crucial for operational decisions like optimizing bike station availability or managing maintenance during adverse weather. For example, it shows precipitation reduces trips within the first six hours, with a noticeable drop in rentals after rainfall.

Long-term analysis reveal sustained or cyclical patterns, which are key for strategic decisions, such as infrastructure planning or policy adjustments. The long-term analysis shows that precipitation's effect on cycling persists throughout the day, emphasizing the need to consider long-term trends for infrastructure changes or policies that accommodate weather variations.

Combining both analyses offers a comprehensive view of weather impacts on bike usage. Short-term analysis optimizes daily operations, while long-term analysis guides strategic decisions for infrastructure and policy. This dual approach helps cities enhance operational efficiency and long-term sustainability in bike-sharing systems.

5.6. Comparison of DiD and causal impact results

The Difference-in-Differences (DiD) analysis, using daily bicycle trip data, suggests that the intervention significantly reduced trips ending in the intervention group compared to the control group (cf. Fig. 9). There is also marginal evidence of a positive effect during the post-intervention period.

The Causal Impact analysis (cf. Fig. 10) shows a significant positive effect, increasing the response variable by 16.8 units (10.7%). The high statistical significance (p = 0.001, posterior probability = 99.89%) indicates the effect is unlikely to be due to chance, strongly supporting the intervention's effectiveness in the post-intervention period.

Both analyses suggest a positive effect of the intervention, though the Causal Impact analysis provides more robust, statistically significant results. While DiD shows some post-intervention effect, its results are less conclusive compared to the Bayesian-based Causal Impact analysis.

6. Discussion

This section discusses the comparison of the results presented in the previous sections. It illustrates the benefits and drawbacks of techniques such as PCMCI, Granger causality, Causal Impact, and DiD by examining cycling patterns and interventions like bridge reopenings. This section also highlights how causal AI has a distinct advantage over

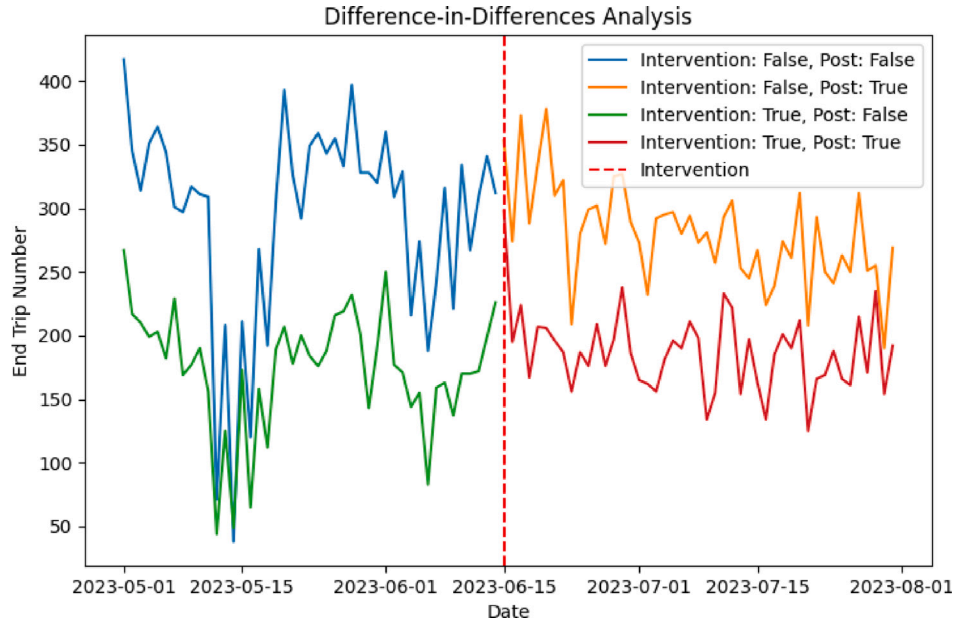

Fig. 9. DiD analysis results showing the intervention's impact on bicycle trips.

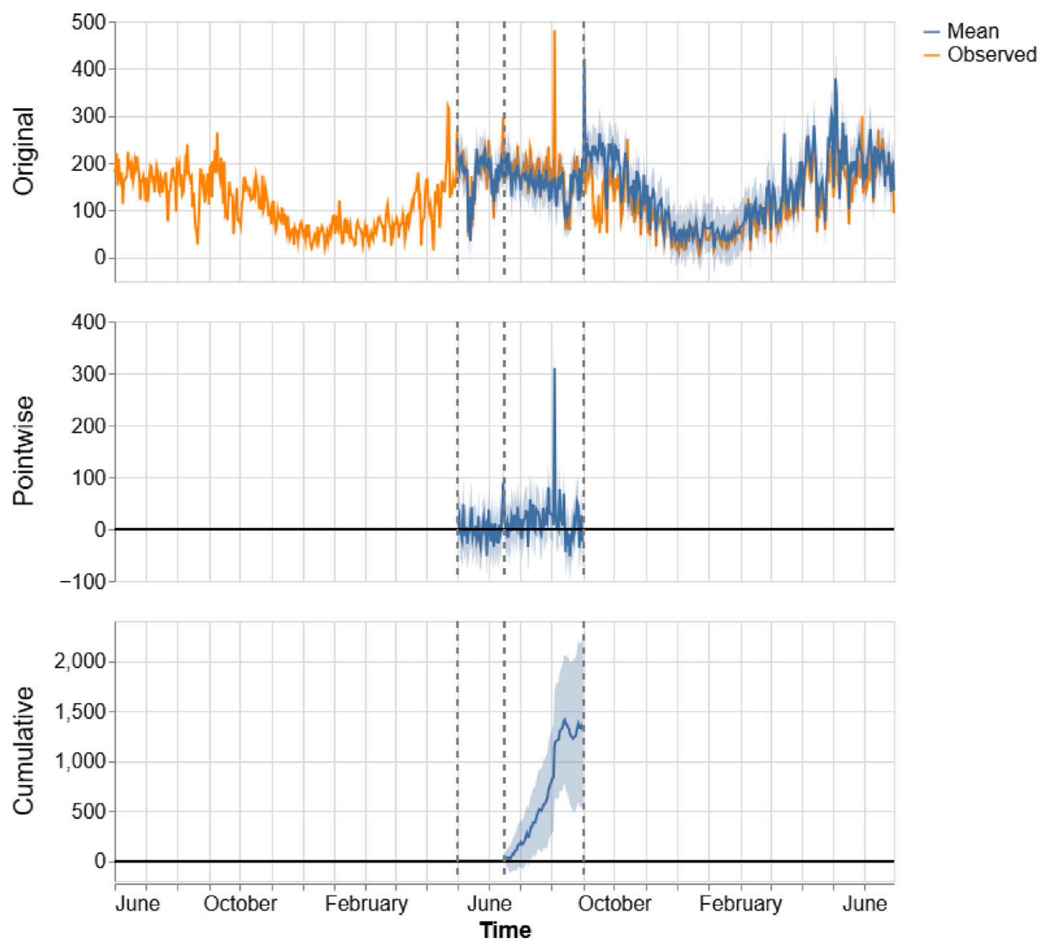

Fig. 10. Causal impact analysis results illustrating the intervention's significant effect.

traditional AI when solving problems like managing temporal dynamics and non-linear interactions and avoiding false correlations. Unlike traditional AI, causal AI aligns with the principles of Judea Pearl's causal hierarchy, modeling real-world cause-and-effect relationships rather than statistical correlations. In addition, the useful uses of causal AI in transportation management and urban planning are discussed, along with the shortcomings of the existing approaches and potential directions for further study. The significance of selecting suitable causal approaches to improve decision-making in sustainable transportation is shown by this thorough examination. This discussion also demonstrates how the findings address the research questions posed in Section 2.

6.1. Comparison of PCMCI and Granger causality

These findings directly address the research questions concerning the limitations of Granger Causality and the advantages of PCMCI. Specifically, they highlight PCMCI's ability to uncover hidden causal dynamics in transport systems, such as the delayed and indirect effects of weather variables on bike usage, which Granger Causality could not detect.

1. **Robustness to high-dimensional data.** PCMCI excels in identifying relationships, as demonstrated in the short-term analysis with *temperature* → *start trip no* (val = 0.218, lag = 1) and *precipitation* → *end trip no* (val = −0.185, lag = 0), both missed by Granger. It captures both direct and delayed effects.
2. **Control of false positives.** In the short-term analysis, PCMCI reduces false positives through conditional independence tests. For example, it detected *relative humidity* → *start trip no* (val = −0.175, lag = 0), which Granger overlooked.
3. **Handling non-linear relationships.** PCMCI captures non-linear effects, such as the influence of *relative humidity* on cycling behavior, missed by Granger in the short-term analysis.
4. **Strength of causation and visualization.** PCMCI quantifies causality strength (*precipitation* → *start trip no*, val = −0.190 in the short-term analysis) and provides clearer visualizations of lag effects.
5. **Temporal structure and lagged effects.** PCMCI identifies lagged relationships, such as *temperature* (lag = 1) → *start trip no*, missed by Granger, emphasizing temporal dynamics in the short-term analysis.

6.2. Comparison of causal impact and DiD

The findings from the Causal Impact analysis directly address the research questions related to DiD and Causal Impact. By demonstrating how infrastructure changes, such as the Chain Bridge reopening, affect bike usage patterns, Causal Impact illustrates its robustness in overcoming the limitations of DiD, including reliance on the parallel trends assumption.

1. **Event-specific insights.** Causal Impact estimated an average increase of 16.8 daily end trips (10.7%) post-bridge reopening, while DiD estimated 26.79 trips.
2. **Dynamic time-series analysis.** Causal Impact better captures post-intervention trends and fluctuations, which DiD overlooks (cf. Fig. 10 vs. Fig. 9).
3. **Uncertainty quantification.** Causal Impact provides confidence intervals, making estimates more robust than DiD's.
4. **Control for baseline trends.** Causal Impact adjusts for pre-intervention trends, avoiding DiD's reliance on the parallel trends assumption.
5. **No parallel trends assumption.** Unlike DiD, Causal Impact does not require perfectly aligned pre-intervention trends, making it more applicable in real-world cases.
6. **Sensitivity to short-term fluctuations.** Causal Impact detects short-term spikes, which DiD aggregates into broader estimates.
7. **Integration of historical data.** Causal Impact leverages historical data for more accurate post-intervention estimates.
8. **Continuous monitoring.** Causal Impact allows ongoing assessment, while DiD offers only a one-time evaluation.

In conclusion, both methods confirm the intervention's success. Despite differences in magnitude, Causal Impact's ability to handle complex time-series trends and provide uncertainty quantification makes it the more robust method in this study.

Table 7
Validation metrics for PCMCI against the gold standard (lag-free evaluation).

Metric	Value
True Positives (TP)	3
False Positives (FP)	2
False Negatives (FN)	0
True Negatives (TN)	10
True Positive Rate (TPR)	1.0
False Positive Rate (FPR)	0.17

6.3. Validation of causal relationships using the gold standard

To evaluate the robustness of the causal relationships identified by our models, we validated them against the gold standard described in Section 4.4. This benchmark, grounded in established domain knowledge and peer-reviewed literature, serves as a reference for assessing the authenticity and robustness of our methods. The validation focuses on key relationships, including temperature, precipitation, and infrastructure changes, as outlined in the gold standard.

To ensure alignment with the simpler structure of the gold standard, we aggregated lagged causal relationships from PCMCI into binary outcomes, indicating the existence or absence of relationships between variable pairs. This adjustment facilitates a direct comparison, as the gold standard does not account for temporal lags.

Key Findings:

- **Temperature and Bike Usage:** PCMCI successfully identified a strong positive relationship (cf. Section 4.6), consistent with the gold standard. This aligns with prior studies indicating that higher temperatures increase cycling activity (Böcker et al., 2013; Liu et al., 2015).
- **Precipitation and Bike Usage:** PCMCI correctly captured the negative effect of rainfall on cycling behavior, reflecting findings from urban transport literature (El-Assi et al., 2017).
- **Bridge Reopening:** The Causal Impact analysis validated the positive effect of the bridge reopening on bike usage (cf. Section 4.7), aligning with prior literature supporting infrastructure's role in enhancing mobility through causal inference (Mölenberg et al., 2019).

While the majority of results aligned with the gold standard, a few divergences were observed:

- **Relative Humidity and Bike Usage:** PCMCI identified a negative effect of relative humidity on cycling behavior, which was not included in the gold standard. This divergence suggests potential latent variables or confounders influencing the results.
- **Weather Interdependencies:** Relationships such as *Temperature* → *Relative Humidity* and *Precipitation* → *Max Wind Speed* were detected. While not directly tied to bike usage, these interdependencies highlight underlying weather patterns that may indirectly affect mobility behaviors.

We assessed PCMCI's accuracy using the following metrics:

- **True Positives (TP):** Matches between PCMCI-detected and gold standard relationships.
- **False Positives (FP):** PCMCI-detected relationships absent in the gold standard.
- **False Negatives (FN):** Gold standard relationships missed by PCMCI.
- **True Negatives (TN):** Correctly identified absence of relationships (see Table 7).

With a True Positive Rate (TPR) of 1.0 and a False Positive Rate (FPR) of 0.17, PCMCI demonstrated strong alignment with the gold standard under the lag-free validation framework. The occasional over-identification of causal links (FP = 2) reflects the complexity of the relationships analyzed but highlights the need for cautious interpretation.

Furthermore, these results validate PCMCI's robustness in identifying meaningful causal links, particularly in operational contexts like sustainable urban mobility policy-making. The observed divergences underscore the need for iterative refinement of the gold standard to incorporate evolving patterns and insights uncovered by causal AI tools.

By integrating algorithmic outputs with expert-driven gold standards, this study establishes a rigorous validation framework for causal discovery. The alignment of key findings with the gold standard reinforces the robustness of the identified causal links between weather patterns, infrastructure interventions, and bike-sharing behaviors. Additionally, causal inference methods demonstrated their utility in validating predefined hypotheses, such as the impact of infrastructure changes.

6.4. Advantages of using causal AI compared to traditional AI

Our approach using causal AI offers several advantages over traditional AI models, which have been introduced in the previous research (Abdellaoui Alaoui & Koumetio Tekouabou, 2021; Albuquerque et al., 2021; Babic et al., 2022; Qian, 2024; Torres et al., 2024; Zeid et al., 2022), particularly in revealing the underlying mechanisms driving changes in bicycle usage. While traditional AI focuses on correlations and prediction, Causal AI provides a deeper understanding of cause-and-effect relationships, which is crucial for data-driven decision-making.

1. **True causal insights.** Traditional AI may predict bike usage increases post-intervention, but Causal AI quantifies the direct impact. For instance, Causal Impact showed a precise daily increase of 16.8 end trips with a 10.7% relative rise, crucial for policy-making and infrastructure planning.
2. **Handling temporal causal dynamics.** Causal AI, unlike traditional models, detects lagged effects. For example, PCMCI revealed that temperature affected *start trip no* with a lag of one period, which traditional AI might overlook by focusing on immediate correlations.
3. **Targeted policy interventions.** Causal AI provides actionable insights, identifying both short-term fluctuations and sustained effects. It enabled policymakers to leverage the 16.8 daily trip increase for resource allocation, insights that traditional models might miss.
4. **Avoiding misleading correlations.** Traditional AI can confuse correlation with causation. For instance, it might falsely associate relative humidity with increased bike usage. Causal AI, on the other hand, identified the true negative relationship (value = −0.175), ensuring more reliable conclusions.
5. **Confidence in decision-making.** Causal AI's ability to quantify uncertainty (e.g., the confidence interval around the 16.8 daily trip increase) enhances the robustness of decisions, offering a reliability that traditional AI often lacks.

In summary, Causal AI provides a more accurate, interpretable framework for analyzing time-series data, revealing the true drivers behind changes in cycling behavior and offering superior insights compared to traditional AI approaches.

6.5. Implications to practice

Urban planners can apply causal discovery on time-series data to understand the causal relationship between weather and bike-sharing platforms. By analyzing how weather impacts cycling patterns, they can create bike-friendly environments and promote active transportation by carefully designing bike lanes, pedestrian pathways, bus stops, and charging stations.

Causal discovery can also be used to promote eco-friendly transportation modes like electric vehicles, walking, and public transit. The insights can guide infrastructure placement to optimize accessibility and encourage greater use of sustainable transport options.

Bike-sharing providers like MOL Bubi can leverage causal discovery to optimize bike availability and maintenance schedules by considering weather forecasts and historical data. Anticipating demand fluctuations helps adjust resource allocation to ensure seamless access to cycling facilities. Furthermore, collaboration with public transport can provide smooth alternatives in poor weather conditions, ensuring a cohesive transport network.

For policy-makers, causal discovery highlights the benefits and challenges of different sustainable transport methods, supporting informed decisions on public investment, subsidies for bike and electric vehicle adoption, and infrastructure development. It also enables the creation of climate-resilient transportation policies by revealing the vulnerabilities of different transport modes to extreme weather.

Causal impact analysis can examine the effect of events like a new bridge on traffic patterns. As demonstrated in this case study, the introduction of a bridge led to increased bicycle usage, allowing policymakers to anticipate similar effects in future infrastructure projects.

6.6. Limitations and future research opportunities

The PCMCI algorithm is a robust tool for causal discovery, but its accuracy depends on identifying confounders. While we included potential confounders in our study, some relevant ones might have been missed, potentially leading to biased results.

The selected parameters – weather variables, operational metrics, and event-specific factors – were sufficient to achieve the study's goals. However, future research could explore additional variables such as socio-economic data, population density, and station-level characteristics. Incorporating these factors may provide a more comprehensive understanding of bike-sharing dynamics and enhance the generalizability of findings to other urban contexts.

As MOL Bubi expands, the current dataset could become outdated. Incorporating data from newly installed stations, as well as past and present values, is essential to maintain the dataset's relevance and accuracy in future analyses.

Future research should also focus on enhancing the gold standard by incorporating domain-specific refinements and exploring trade-offs between sensitivity and specificity in causal discovery. These improvements could provide greater robustness in identifying causal links while addressing the evolving complexity of urban mobility systems.

Although our study focuses primarily on Bubi stations, a more comprehensive understanding of urban mobility also requires analyzing other sustainable transport modes. We incorporated domain knowledge, such as the influence of weather on transportation, to improve our results. However, further collaboration with experts is needed to refine confounder identification and analysis.

Causal impact analysis also has limitations, including selection bias and unobserved variables. For example, intervention and control groups must be well-chosen to ensure only the intervention group is affected by the event. Additionally, unobserved factors, like a summer school break near universities, could impact the control group without affecting the intervention group.

However, implementing causal AI in real-world scenarios comes with its own set of challenges. Several key challenges in causal AI are highlighted in the literature (Vallverdú, 2024). Beyond distinguishing causation from mere correlation, causal AI requires extensive domain knowledge, which can be incomplete or poorly integrated into models, leading to inaccuracies. Additionally, assumptions underlying causal models can introduce errors if they do not accurately represent the true causal mechanisms. Furthermore, integrating causal AI with generative models adds complexity and raises ethical considerations. Addressing these challenges requires balancing domain expertise with accurate model assumptions to improve causal inference and ensure successful application.

7. Conclusions

This study integrated causal AI methods into time-series analysis to enhance our understanding of causal relationships in sustainable transportation behavior.

Our findings demonstrate the potential of causal AI to transform urban transportation planning. Uncovering hidden causal relationships in bike-sharing data offers actionable insights for policymakers looking to optimize transport systems for sustainability. This directly addresses the identified research gap, particularly the limited application of causal AI in practical contexts and the scarcity of case studies demonstrating its utility. Causal AI emerges as a powerful tool for developing data-driven solutions that address modern city challenges.

We analyzed both short- and long-term causal relationships, employing resampling and filtering techniques to reveal nuanced insights. Our results show that weather impacts cycling behavior across different time scales, highlighting the importance of time-sensitive interventions. We also applied causal impact analysis to estimate the effects of the bridge reopening intervention, which we compared with Difference-in-Differences (DiD). Our findings showed that causal impact provided more precise and reliable insights, overcoming challenges in data quality and interpretability noted in previous literature.

Causal discovery and inference in time-series data has significant implications for transportation planning, service optimization, and policymaking. Urban planners can design bike-friendly environments by incorporating weather-sensitive strategies into cycling infrastructure and promoting eco-friendly transportation. Providers like MOL Bubi can dynamically adjust bike availability based on predicted demand patterns derived from weather forecasts and historical data. Policymakers can use these insights to develop climate-resilient policies and encourage sustainable transport initiatives. By addressing these challenges, this study bridges the gap between theoretical advancements in causal AI and their practical applications in urban mobility.

In conclusion, causal discovery and causal effect estimation have provided valuable insights into sustainable transport behavior. By identifying causal relationships rather than correlations, we are better equipped to make informed decisions, fostering a more sustainable and resilient urban transport system. The significance of this study lies in its demonstration of how causal AI can address real-world urban mobility challenges, offering a replicable framework for other cities to optimize transportation systems and reduce environmental impacts.

While this study focused on key factors influencing bike-sharing dynamics, future research could explore additional contextual variables and multimodal transport systems, as highlighted in our discussion of limitations. These efforts would further refine causal models and expand their applicability to broader urban mobility strategies.

This study provides a foundation for replicating causal AI frameworks in diverse urban contexts, bridging the gap between data-driven decision-making and actionable urban mobility strategies.

Nomenclature

Symbol	Definition
CI	Conditional Independence, used to determine causal relationships.
$Cov(X,Y)$	Covariance between variables X and Y.
DiD	Difference-in-Differences, a causal effect estimation method.
N	Number of observations in the dataset.
$PCMCI$	Peter and Clark Momentary Conditional Independence, a causal discovery algorithm.
$P(X\|Z)$	Probability of X given Z, used in conditional independence tests.
$r_{X\to Y\|Z}$	Partial correlation coefficient, conditioned on Z.
τ_{max}	Maximum lag in time-series analysis.
Y_{cf}	Counterfactual prediction in Causal Impact analysis.
Y_{obs}	Observed outcome in time-series data.
δ_t	Causal impact at time t, computed as $Y_{obs} - Y_{cf}$.

CRediT authorship contribution statement

Tamas Fekete: Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Girum Mengistu:** Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Hendro Wicaksono:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to improve language. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

A.1. Results for short-term Granger analysis

See Fig. A.11.

A.2. Results for long-term Granger analysis

See Fig. A.12.

A.3. Detailed results for short-term PCMCI analysis

These tables show the significant links identified by the PCMCI algorithm, including the p-values and the strength of the causal relationships (represented by the "Value" column) at various lags. The tables include the following columns:

- **Variable:** The name of the variable that has a causal relationship with `end_trip_no`.
- **Lag:** The time lag at which the causal relationship is observed. A negative lag indicates that the variable at a previous time point affects `end_trip_no`.
- **p-value:** The probability value indicates the significance of the causal relationship. A lower p-value suggests a more significant relationship.
- **Value:** The strength and direction of the causal relationship. A positive value indicates a positive relationship, while a negative value indicates a negative relationship.

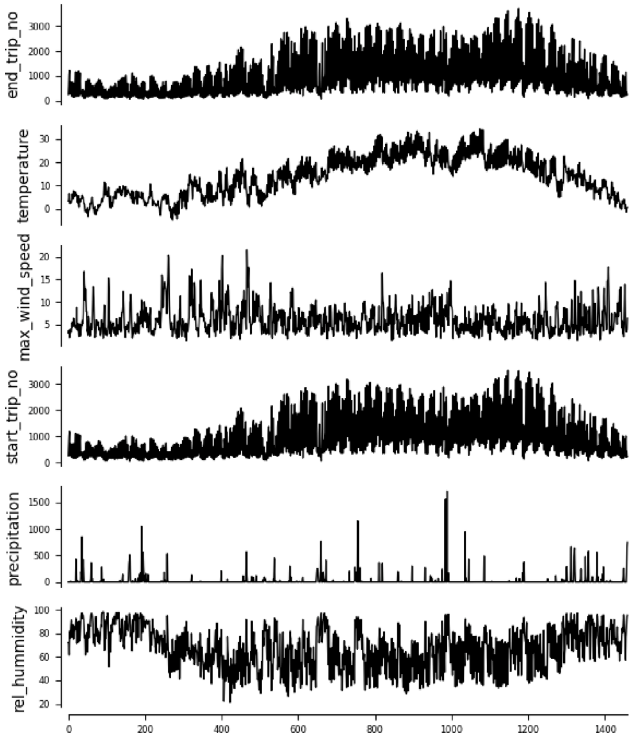

Fig. A.11. Time-series graph for short-term analysis.

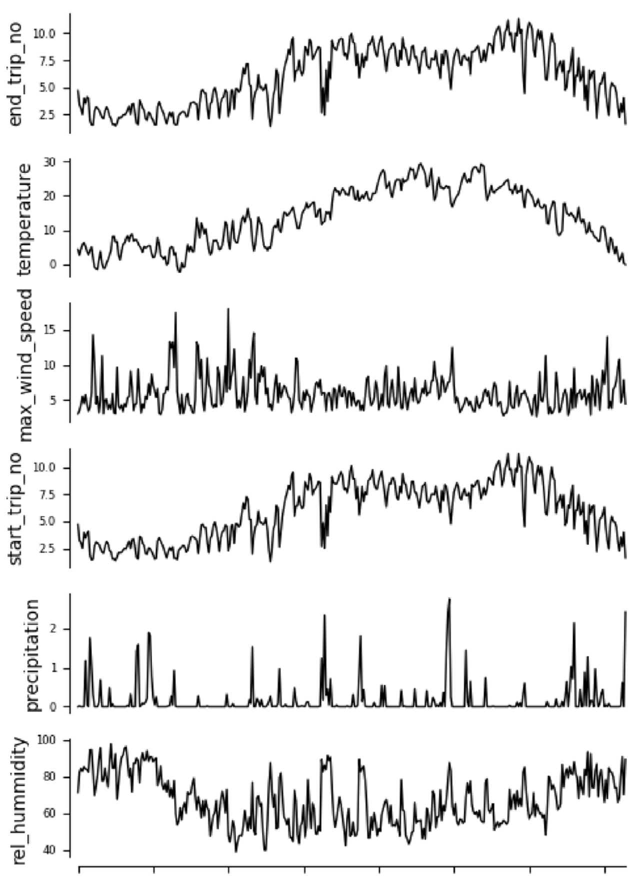

Fig. A.12. Time series graph for long-term analysis.

Table A.8
Links for variable `end_trip_no`.

Variable	Lag	p-value	Value
start_trip_no	0	0.00000	0.985
end_trip_no	−2	0.00000	0.306
temperature	−1	0.00000	0.225
temperature	−2	0.00000	−0.206
precipitation	0	0.00000	−0.185
precipitation	−1	0.00000	−0.167
start_trip_no	−2	0.00000	0.165
rel_hummidity	−1	0.00000	−0.162
temperature	0	0.00000	0.161
rel_hummidity	0	0.00000	−0.154
start_trip_no	−3	0.00000	−0.143
max_wind_speed	−5	0.00000	0.138
end_trip_no	−3	0.00002	−0.127
precipitation	−5	0.00003	0.124
max_wind_speed	0	0.00001	−0.121
end_trip_no	−4	0.00014	0.115
start_trip_no	−6	0.00020	−0.113
max_wind_speed	−1	0.00024	−0.111
end_trip_no	−8	0.00038	0.109
rel_hummidity	−4	0.00104	0.102
precipitation	−4	0.00265	0.094
start_trip_no	−10	0.00881	−0.084
rel_hummidity	−3	0.00933	0.084
rel_hummidity	−6	0.01510	−0.079
rel_hummidity	−10	0.01889	−0.077
end_trip_no	−1	0.02540	0.074
end_trip_no	−9	0.03423	0.072

Table A.9
Links for variable `temperature`.

Variable	Lag	p-value	Value
rel_hummidity	0	0.00000	−0.675
temperature	−1	0.00000	0.631
temperature	−2	0.00000	−0.230
temperature	−3	0.00000	0.192
start_trip_no	0	0.00000	0.188
end_trip_no	0	0.00000	0.161
precipitation	0	0.00000	−0.159
temperature	−4	0.00000	0.159
temperature	−9	0.00000	−0.155
temperature	−5	0.00000	−0.143
max_wind_speed	−1	0.00001	−0.133
rel_hummidity	−1	0.00002	0.127
rel_hummidity	−8	0.00004	−0.123
rel_hummidity	−2	0.00020	−0.113
rel_hummidity	−4	0.00056	−0.106
temperature	−8	0.00262	0.095
precipitation	−2	0.00564	−0.088
temperature	−7	0.02540	−0.074
precipitation	−7	0.04836	−0.068
max_wind_speed	0	0.01464	0.065

Table A.8 presents the significant links for the variable `end_trip_no`.

Table A.9 presents the significant links for the variable `temperature`.

Table A.10 presents the significant links for the variable `start_trip_no`.

Table A.11 presents the significant links for the variable `precipitation`.

Table A.12 presents the significant links for the variable `rel_hummidity`.

A.4. Detailed results for long-term PCMCI analysis

The following tables provide detailed long-term PCMCI analysis results for each variable of interest. Each table shows the links, lags, p-values, and values between a specific variable and others, highlighting their relationships.

Table A.10
Links for variable `start_trip_no`.

Variable	Lag	p-value	Value
end_trip_no	0	0.00000	0.985
start_trip_no	−4	0.00000	0.488
temperature	−1	0.00000	0.218
temperature	−2	0.00000	−0.202
precipitation	0	0.00000	−0.190
temperature	0	0.00000	0.188
rel_hummidity	0	0.00000	−0.175
rel_hummidity	−4	0.00000	0.168
rel_hummidity	−1	0.00000	−0.166
precipitation	−1	0.00000	−0.150
start_trip_no	−2	0.00001	0.129
max_wind_speed	0	0.00000	−0.125
start_trip_no	−6	0.00003	−0.124
precipitation	−5	0.00007	0.119
end_trip_no	−6	0.00007	−0.119
rel_hummidity	−10	0.00036	−0.109
start_trip_no	−10	0.00097	−0.102
max_wind_speed	−1	0.00101	−0.102
end_trip_no	−8	0.00113	0.101
start_trip_no	−3	0.00157	−0.098
temperature	−5	0.00157	−0.098
precipitation	−4	0.00428	0.090
max_wind_speed	−5	0.00769	0.086
end_trip_no	−9	0.01510	0.079
end_trip_no	−2	0.02540	−0.074
start_trip_no	−9	0.02540	0.074
rel_hummidity	−5	0.04420	0.069
end_trip_no	−3	0.04836	−0.068

Table A.11
Links for variable `precipitation`.

Variable	Lag	p-value	Value
rel_hummidity	0	0.00000	0.312
precipitation	−1	0.00000	0.229
start_trip_no	0	0.00000	−0.190
end_trip_no	0	0.00000	−0.185
precipitation	−4	0.00000	0.179
max_wind_speed	0	0.00000	0.166
temperature	0	0.00000	−0.159
rel_hummidity	−8	0.00797	0.085
rel_hummidity	−1	0.04420	0.069

Table A.12
Links for variable `rel_hummidity`.

Variable	Lag	p-value	Value
temperature	0	0.00000	−0.675
rel_hummidity	−1	0.00000	0.547
precipitation	0	0.00000	0.312
max_wind_speed	0	0.00000	−0.225
rel_hummidity	−2	0.00000	−0.213
start_trip_no	0	0.00000	−0.175
end_trip_no	0	0.00000	−0.154
temperature	−4	0.00000	−0.142
rel_hummidity	−3	0.00001	0.131
temperature	−8	0.00093	−0.103
start_trip_no	−1	0.00265	0.094
temperature	−1	0.00341	0.092
end_trip_no	−1	0.00419	0.091
rel_hummidity	−5	0.00787	−0.085
temperature	−9	0.00968	0.083
max_wind_speed	−1	0.01048	−0.082
rel_hummidity	−4	0.01806	0.078
temperature	−3	0.04299	−0.069

Table A.13 presents the significant links for the variable `end_trip_no`.

Table A.14 presents the significant links for the variable `temperature`.

Table A.15 presents the significant links for the variable `max_wind_speed`.

Table A.13
Links for variable `end_trip_no`.

Variable	Lag	p-value	Value
start_trip_no	0	0.00000	0.997
rel_hummidity	0	0.00000	−0.575
temperature	0	0.00000	0.548
precipitation	0	0.00000	−0.480
max_wind_speed	0	0.00001	−0.251
temperature	−14	0.02804	−0.200
precipitation	−2	0.03182	−0.194

Table A.14
Links for variable `temperature`.

Variable	Lag	p-value	Value
temperature	−1	0.00000	0.712
end_trip_no	0	0.00000	0.548
start_trip_no	0	0.00000	0.539
rel_hummidity	0	0.00000	−0.365
max_wind_speed	−1	0.00000	−0.328
temperature	−2	0.00350	−0.229
precipitation	0	0.00087	−0.185
max_wind_speed	0	0.02176	−0.130

Table A.15
Links for variable `max_wind_speed`.

Variable	Lag	p-value	Value
max_wind_speed	−1	0.00000	0.404
end_trip_no	0	0.00001	−0.251
start_trip_no	0	0.00001	−0.250
rel_hummidity	0	0.00150	−0.178
temperature	0	0.02176	−0.130

Table A.16
Links for variable `start_trip_no`.

Variable	Lag	p-value	Value
end_trip_no	0	0.00000	0.997
rel_hummidity	0	0.00000	−0.564
temperature	0	0.00000	0.539
precipitation	0	0.00000	−0.479
max_wind_speed	0	0.00001	−0.250
temperature	−14	0.02600	−0.203

Table A.17
Links for variable `precipitation`.

Variable	Lag	p-value	Value
rel_hummidity	0	0.00000	0.490
end_trip_no	0	0.00000	−0.480
start_trip_no	0	0.00000	−0.479
precipitation	−1	0.00005	0.273
temperature	0	0.00087	−0.185

Table A.18
Links for variable `rel_hummidity`.

Variable	Lag	p-value	Value
end_trip_no	0	0.00000	−0.575
start_trip_no	0	0.00000	−0.564
precipitation	0	0.00000	0.490
rel_hummidity	−1	0.00000	0.456
temperature	0	0.00000	−0.365
max_wind_speed	0	0.00150	−0.178

Table A.16 presents the significant links for the variable `start_trip_no`.

Table A.17 presents the significant links for the variable `precipitation`.

Table A.18 presents the significant links for the variable `rel_hummidity`.

Data availability

Data will be made available on request.

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